Week 5
Fully preprocessed data (for DL model)
Deep Learning model performance (iteration 1); discussion of what went wrong and how to fix it

● Ensure your data is preprocessed correctly for the deep learning model you're using
Example: for image data, rotation, progressive resizing, mixup
We reorganized the data so that each column represents an individual household and each row represents a unique time step.
In order to save on memory space, we also tried several data merging methods, such as taking the average of usage over all households to serve as a representative household and randomly selecting 10 households to train the model on a mini batch of samples.
We also considered clustering different households based on their mean energy consumption and standard deviation.
Further preprocessing might be required to keep computing time accessible; this will be explored as needed.
In order to prevent data leakage, we shall select 20% of the households at random as the test set, which will not be touched until all models are finalized.

● Put your preprocessed data through the deep learning architecture.
We succeeded in passing our time series data for several households through the LSTM model. Overall training time took approximately 2 hours.

● List the performance metrics and any useful diagnostic figures
Our main performance metric is the Average (over multiple households) MSE of predictions over a fixed time .
Currently, the MSE on our testing data is 0.005949, and the average MSE on 5 unseen test houses is 0.0071 for our mini LSTM baseline model built upon 10 randomly selected households.

● Describe where the model worked well and where it didn't work well
We can observe an order of magnitude improvement of MSE over baseline models.
We are still struggling with the magnitude of the dataset, in that our current model does not incorporate all households with all of their granulated time steps (every 30 minutes). We need to figure out a way to incorporate more households which should improve the model's predictive power.